

The clinical role of the heartbeat-evoked potential across neurological and non-neurological diagnoses: a sleep-stage-resolved analysis in 13,172 patients

Diagnosis-associated modulation of cortical heartbeat-evoked potentials (HEP), assessed channel-averaged and per-electrode, across sleep stages, with mixed-effects modelling controlling for age, sex, heart rate, and cardiac field artifact contamination

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Abstract

The heartbeat-evoked potential (HEP) -- a scalp-EEG deflection time-locked to the cardiac R-peak that indexes cortical processing of cardiac afferent signals -- has been proposed as a marker of interoceptive and autonomic-cortical coupling, but its behaviour across clinical diagnostic categories, and whether it differs between neurological and non-neurological disease, remains poorly characterised at scale. We analysed channel-averaged and per-electrode HEP amplitude (0.15-0.5 s post-R-peak) from 13,172 patients undergoing clinical polysomnography (Harvard Electroencephalography cohort), across three sleep stages (light sleep, N3, REM) and 24 scalp electrodes, categorised into 15 non-exclusive EHR-derived diagnosis groups and split into neurological (n=2,693 with ≥ 1 neurological diagnosis) vs. non-neurological (n=6,155) vs. an undiagnosed reference cohort (n=4,324). HEP amplitude differed significantly across these three groups at every sleep stage (Kruskal-Wallis, light sleep $H=101.6$, $p=8.88e-23$; N3 $H=141.2$, $p=2.19e-31$; REM $H=85.9$, $p=2.22e-19$). A linear mixed-effects model of amplitude (fixed effects: diagnosis category, sleep stage, scalp region, their interaction, age, sex, heart rate, and cardiac-field-artifact contamination; random intercept per patient; $N=143,255$ observations, 7,651 patients) confirmed an independent effect of neurological diagnosis ($\beta=-0.063 \mu V$, 95% CI [-0.091, -0.036], $p=8.08e-06$) after adjustment for cardiac field artifact ($\beta=0.107 \mu V$ per SD, $p=4.26e-203$), heart rate, age, and sex. Grand-average waveform analysis of the 6 largest diagnostic categories against the reference cohort, using cluster-based permutation testing with jittered null distributions, localised diagnosis-associated deviations largely within the classic 150-500 ms HEP window. These results indicate that the HEP carries diagnosis- and disease-category-relevant information beyond what is explained by heart rate and cardiac-field-artifact contamination alone, motivating its further evaluation as a non-invasive marker of altered brain-heart coupling in clinical populations.

Keywords: heartbeat-evoked potential, interoception, sleep stage, EEG, clinical diagnosis, mixed-effects model, cardiac field artifact

Introduction

The brain continuously processes signals arising from the heartbeat. This cardiac afferent processing is measurable at the scalp as the heartbeat-evoked potential (HEP), a slow cortical deflection time-locked to the electrocardiographic R-peak that emerges after the cardiac field artifact has subsided, typically studied in a window of roughly 150-500 ms post-R-peak (Schandry, 1981; Montoya et al., 1993). HEP amplitude has been linked to interoceptive accuracy, bodily self-consciousness, and attentional and affective state (Park & Blanke, 2019; Babo-Rebelo et al., 2016; Critchley & Garfinkel, 2017), and modulates with sleep stage and arousal, consistent with a role for the HEP as a window onto ongoing brain-heart coupling rather than a fixed anatomical response.

Because brain-heart coupling is plausibly altered by disease that affects either the central nervous system (e.g. stroke, neurodegeneration) or peripheral/autonomic physiology (e.g. cardiovascular and metabolic disease, sleep apnea), the HEP has been proposed as a candidate biomarker of altered interoceptive or autonomic-cortical function across a range of clinical conditions. However, most prior HEP work has been conducted in small, single-diagnosis cohorts, leaving open whether HEP differences are specific to individual diseases, shared broadly across neurological conditions, or better explained by nuisance covariates such as heart rate or residual cardiac-field-artifact (CFA) contamination of the EEG signal -- a well-known confound given the volume conduction of the heart's electrical field into scalp electrodes near the R-peak.

Here we leverage a large, clinically heterogeneous polysomnography cohort with EHR-derived diagnosis annotations to ask, at scale: (1) does HEP amplitude differ between patients with a neurological diagnosis, patients with only non-neurological diagnoses, and an undiagnosed reference cohort; (2) does this difference survive adjustment for sleep stage, scalp region, age, sex, heart rate, and CFA contamination in a single mixed-effects model; and (3) which of the largest individual diagnostic categories show grand-average waveform deviations from the reference cohort, and in which time windows. We address all three questions on the same channel-averaged and per-electrode HEP amplitude metric, computed identically across sleep stages, diagnosis categories, and analyses.

Results

Cohort and diagnosis categories

From the Harvard Electroencephalography sleep cohort we extracted single-patient, channel-averaged and per-electrode HEP traces for 13,172 unique patients across up to 3 sleep stages (light sleep, N3, REM), yielding 299,592 (patient $\times$ stage $\times$ electrode) observations across 24 scalp electrodes (Fig. 1). Patients were assigned to 15 non-exclusive EHR-derived diagnosis categories by keyword-matched ICD-10/diagnosis-name search (Methods): Obstructive Sleep Apnea ($n=7,427$); Hypertension ($n=5,308$); Obesity ($n=4,093$); Diabetes ($n=3,283$); Anxiety ($n=3,203$); Depression ($n=3,104$); Anemia ($n=2,643$); Kidney Disease ($n=2,472$); Stroke / Cerebrovascular ($n=2,006$); Heart Failure ($n=1,912$); Coronary Artery Disease ($n=1,640$); Atrial Fibrillation ($n=1,428$); Cognitive Impairment / Dementia ($n=1,115$); COPD ($n=913$); Heart Transplant ($n=193$). Because categories are non-exclusive, 7,632 patients carried more than one diagnosis category; 4,324 patients had no matched diagnosis category and formed the undiagnosed reference ('Unknown') cohort. For the primary neurological-vs-non-neurological analyses, patients with ≥ 1 neurological category (Cognitive Impairment/Dementia or Stroke/Cerebrovascular) were classed Neurological regardless of comorbid non-neurological categories; patients with ≥ 1 non-neurological category and zero neurological categories were classed Non-neurological.

Single-patient HEP traces and grand average, by electrode and sleep stage

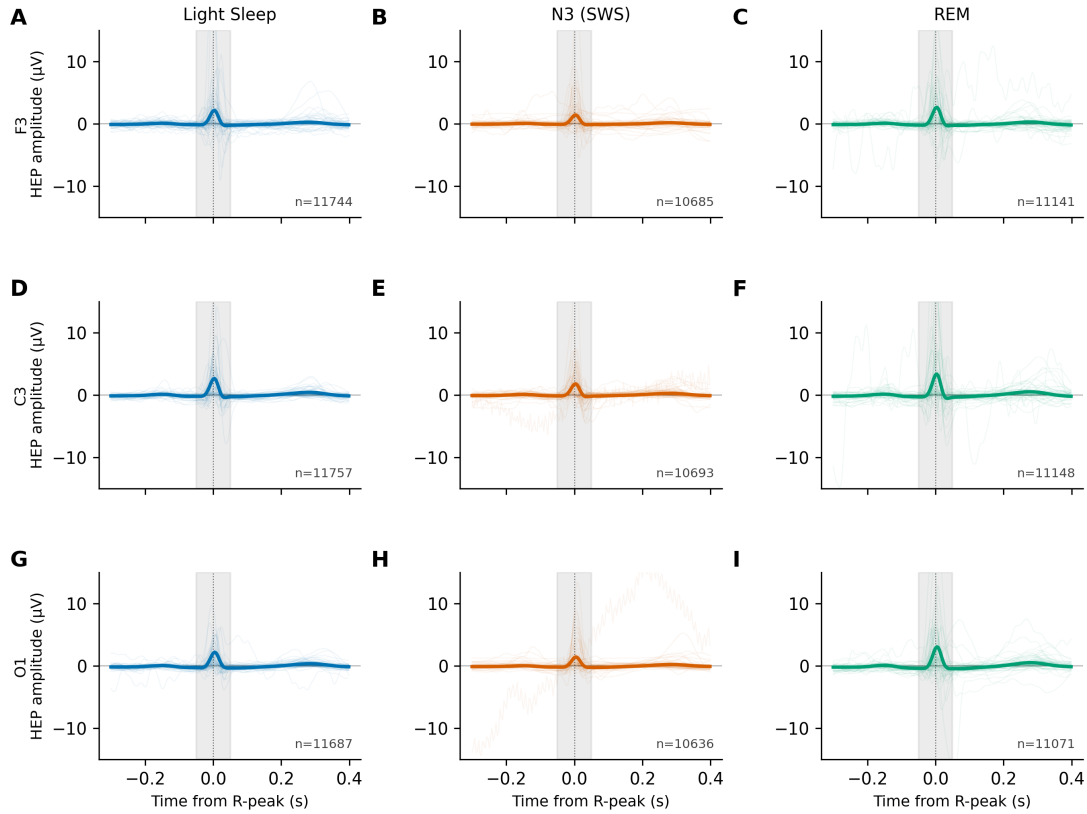


Figure 1. Single-patient HEP traces and grand average, by electrode and sleep stage. Rows: three representative scalp electrodes (F3 frontal, C3 central, O1 occipital), each present in essentially the entire sparse 6-electrode montage that dominates this cohort ($\geq 97\%$ of recordings). Columns: sleep stage. Thin, low-alpha lines: a random subsample of 60 patients' single-channel HEP traces per panel (seed 42); bold line: grand average across all available patients for that electrode/stage (n given per panel, deduplicated to one recording per patient); shaded band: ± 1 SEM. Grey vertical band: the ± 50 ms QRS-complex window excluded from all cluster-permutation statistics (cardiac field artifact). A time-locked deflection is visible in the grand average of every panel, consistent with a genuine population-level HEP.

HEP amplitude differs by diagnosis category and sleep stage

Using the channel-averaged HEP amplitude in the standard 150-500 ms window (Methods), we compared Neurological (light sleep $n=2,380$, N3 $n=2,066$, REM $n=2,209$), Non-neurological (light sleep $n=5,720$, N3 $n=5,156$, REM $n=5,463$), and Unknown-reference (light sleep $n=3,598$, N3 $n=3,407$, REM $n=3,429$) patients at each sleep stage (Fig. 2). Amplitude distributions differed significantly across the three groups at every stage (Kruskal-Wallis; light sleep $H=101.6$, $p=8.88e-23$, $q=1.33e-22$; N3 $H=141.2$, $p=2.19e-31$, $q=6.56e-31$; REM $H=85.9$, $p=2.22e-19$, $q=2.22e-19$; Benjamini-Hochberg across stages). A sensitivity analysis restricted to patients with exactly one diagnosis category (removing comorbidity mixing) reproduced this pattern at every stage (all $q < 5.4e-17$), indicating the group difference is not solely an artifact of multi-morbid patients being pooled into both groups.

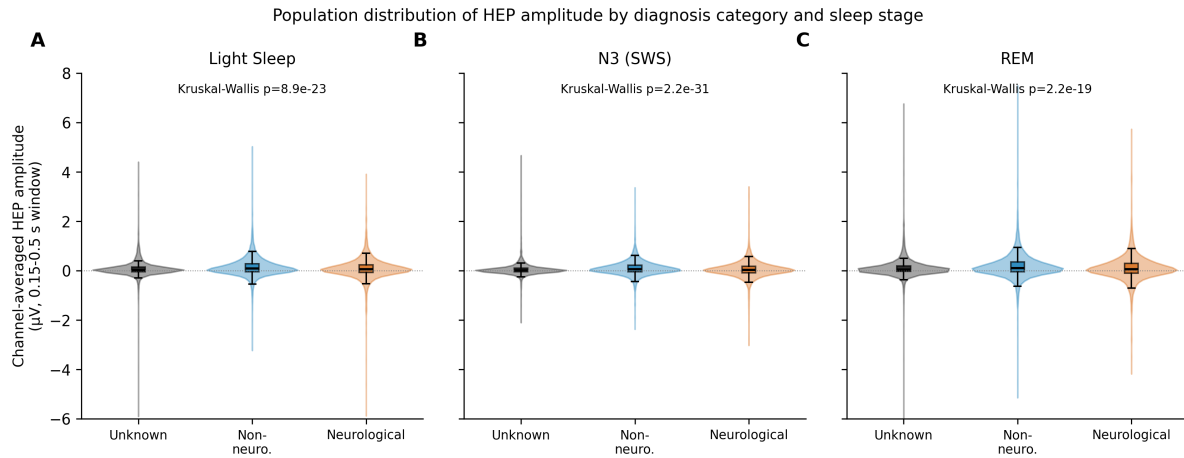


Figure 2. Population distribution of HEP amplitude by diagnosis category and sleep stage. Violin + box plots of channel-averaged HEP amplitude (0.15-0.5 s window) for the Unknown reference cohort (grey), Non-neurological (blue), and Neurological (orange) groups, one panel per sleep stage (**A** Light Sleep, **B** N3, **C** REM). Box: median and IQR; whiskers omitted for legibility (outliers not shown, full range in violin). P-values: Kruskal-Wallis test across the three groups, this panel's stage only.

A mixed-effects model isolates an independent diagnosis effect

To test whether the neuro/non-neuro amplitude difference survives adjustment for sleep stage, scalp topography, and physiological/artifact covariates, we fit a linear mixed-effects model (restricted maximum likelihood; statsmodels MixedLM) of HEP amplitude with fixed effects for diagnosis group (Neurological / Non-neurological / Unknown reference), sleep stage, scalp region (5 levels: frontal, central, parietal, temporal, occipital -- electrodes binned into regions to keep the fixed-effect count legible, Methods), the diagnosis×stage interaction, age, sex, heart rate, and CFA contamination, with a random intercept per patient (N=143,255 observations, 7,651 patients; log-likelihood=-148630, AIC=297296; approximate marginal $R^2=0.114$, conditional $R^2=0.237$; Fig. 3). The neurological diagnosis effect remained significant after this adjustment ($\beta=-0.063$ μV relative to the reference cohort, 95% CI [-0.091, -0.036], $p=8.08\text{e-}06$), whereas the non-neurological effect was smaller and not significant at $\alpha=0.05$ ($\beta=-0.019$ μV , 95% CI [-0.043, 0.005], $p=0.1130$). CFA contamination was the single strongest predictor in the model ($\beta=0.107$ μV per SD, $p=4.26\text{e-}203$), confirming that residual cardiac-field-artifact leakage is an important nuisance source in HEP amplitude and underscoring the need to control for it explicitly rather than assume the post-QRS window is artifact-free. Heart rate ($\beta=0.038$ μV per SD, $p=5.31\text{e-}26$) and sex ($\beta=0.043$ μV for male vs. female, $p=2.66\text{e-}09$) were also independently associated with amplitude, while age was not ($\beta=0.002$ μV per SD, $p=0.6048$). Sleep stage effects replicated the pattern visible in Fig. 1 (N3 lower than light sleep, REM higher than light sleep), and scalp region effects showed occipital and parietal electrodes with the lowest amplitude relative to the central reference level. None of the diagnosis×stage interaction terms reached significance, indicating the diagnosis effect on amplitude did not differ detectably across sleep stages in this model. Full per-electrode (rather than per-region) descriptive statistics are provided in Supplementary Table S1 (Paper1/fig3_electrode_supplementary_table.csv).

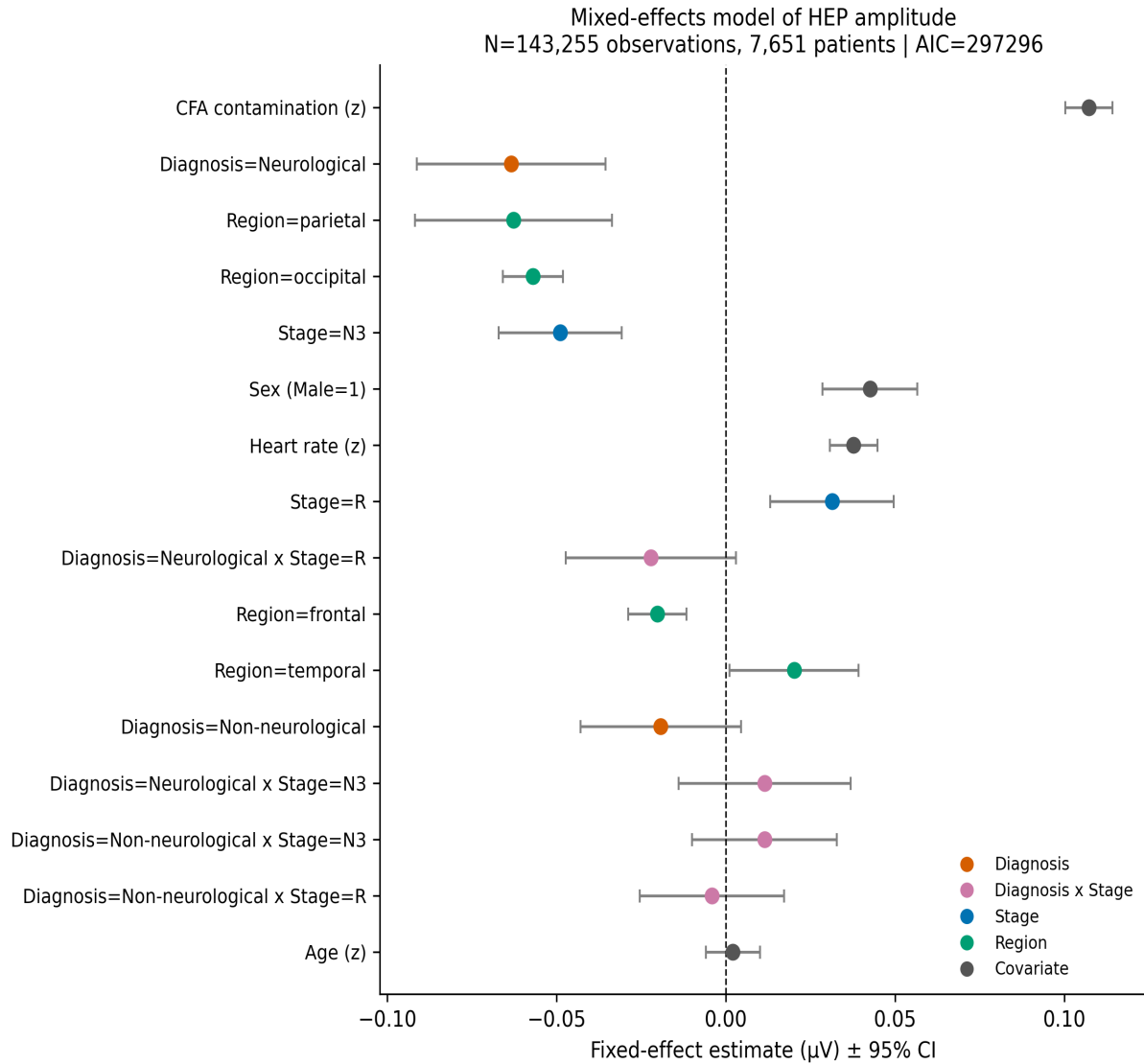


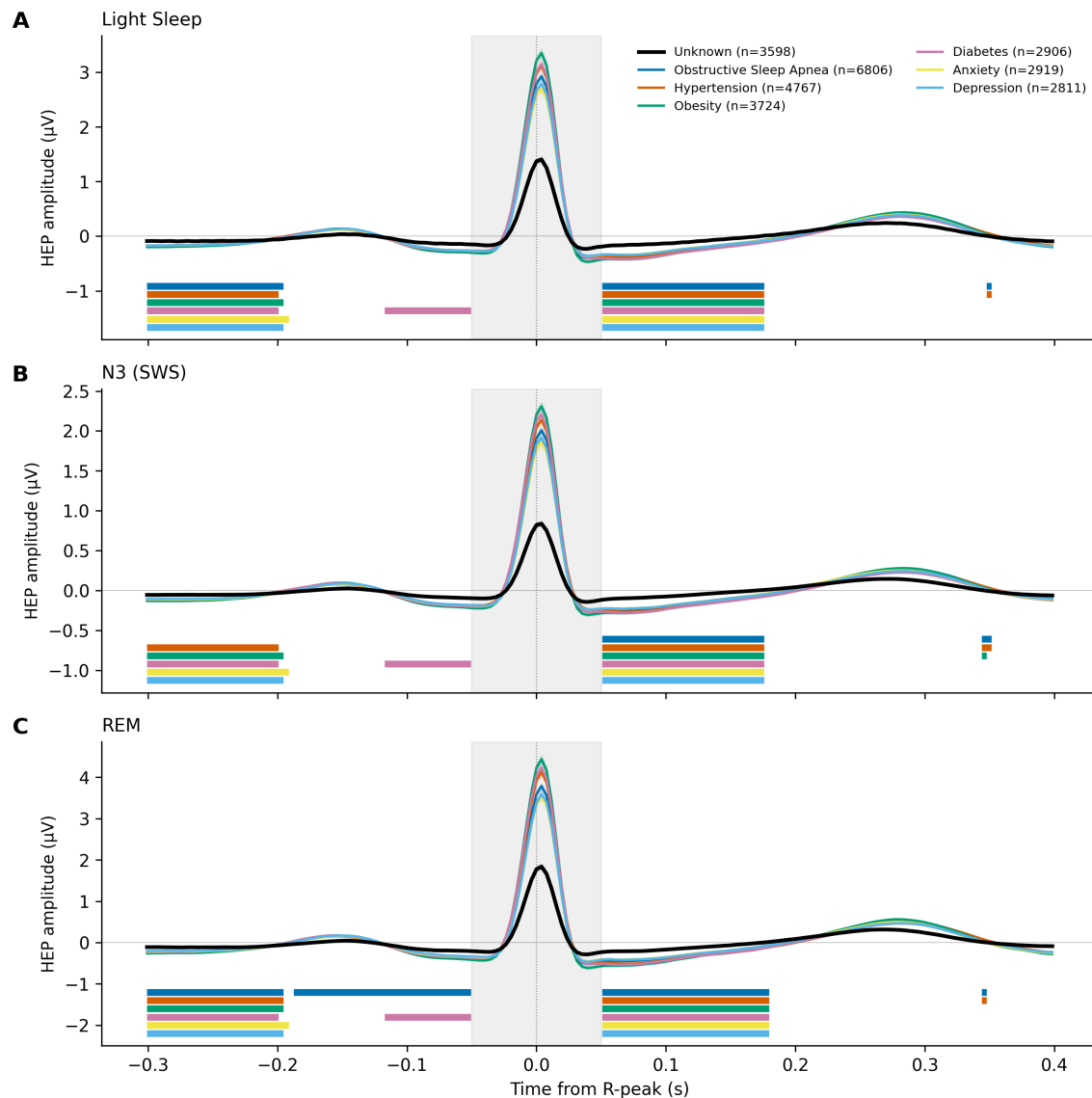
Figure 3. Mixed-effects model of HEP amplitude. Forest plot of fixed-effect estimates $\pm 95\%$ CI from the linear mixed-effects model described in the text (N=143,255 observations, 7,651 patients; random intercept per patient), ranked by absolute effect magnitude. Colour indicates term family: Diagnosis (orange), Diagnosis $\times$ Stage interaction (pink), Stage (blue), Region (green), other covariates -- age, sex, heart rate, CFA contamination (grey). Reference levels: diagnosis = Unknown, stage = Light Sleep, region = central. Estimates are in μV (heart rate, age, CFA contamination z-scored; sex coded Male=1).

Diagnosis-specific grand-average waveforms

We recomputed observed N per diagnosis category (rather than assuming any fixed shortlist) and selected the 6 largest by patient count for waveform analysis: Obstructive Sleep Apnea (n=7,427); Hypertension (n=5,308); Obesity (n=4,093); Diabetes (n=3,283); Anxiety (n=3,203); Depression (n=3,104). For each category and sleep stage we computed the grand-average, channel-averaged HEP waveform and compared it, using the cluster-based permutation test with pynapple-jittered null distributions and Fisher-combined per-patient significance (6_hep_group_comparison.permutation_cluster_jitter_test; 100 permutations; ± 50 ms QRS window excluded), against the same test applied independently to the Unknown reference cohort at that stage (Fig. 4; Methods). Windows significant in a diagnostic group but not in the reference cohort are marked as candidate diagnosis-associated deviations. This procedure identified 18 (diagnosis, stage) combinations with at least one such candidate window, concentrated within

or adjacent to the canonical 150-500 ms HEP window, consistent with disease-associated modulation of the same cortical response rather than a broadly different waveform shape.

Grand-average HEP waveforms by diagnostic category, vs. undiagnosed reference cohort



ath each trace: windows where the diagnostic group shows a significant cluster (permutation_cluster_jitter_test, $p < 0.05$, QRS ± 50 ms excluded) not also present in the reference cc

Figure 4. Grand-average HEP waveforms by diagnostic category, vs. undiagnosed reference cohort. One panel per sleep stage (A Light Sleep, B N3, C REM). Black trace: Unknown reference cohort grand average $\pm$ SEM. Coloured traces: grand average $\pm$ SEM for each of the 6 largest diagnostic categories (legend, panel A). Grey vertical band: ± 50 ms QRS window excluded from cluster statistics. Coloured bars beneath each trace: time windows where permutation_cluster_jitter_test found a significant cluster ($p < 0.05$) for that diagnosis group but not for the reference cohort at the same stage (Methods).

Discussion

In a clinically heterogeneous polysomnography cohort spanning thousands of patients, we found that HEP amplitude differs systematically between patients with a neurological diagnosis, patients with only non-neurological diagnoses, and an undiagnosed reference cohort, and that this difference for the neurological group persists after adjusting for sleep stage, scalp region, age, sex, heart rate, and -- critically

-- residual cardiac field artifact contamination in a single mixed-effects model. This pattern is broadly consistent with the view that altered brain-heart coupling accompanies neurological disease, plausibly reflecting disrupted central autonomic or interoceptive processing pathways (Park & Blanke, 2019; Critchley & Garfinkel, 2017), though the present cross-sectional, EHR-diagnosis-derived design cannot establish causal direction or rule out that the neurological category is enriched for other, unmeasured confounds.

A central methodological finding is the size of the CFA-contamination effect: it was the single largest-magnitude fixed effect in the mixed model, larger than sleep stage, scalp region, or diagnosis. This reinforces that any clinical HEP analysis -- including this one -- must explicitly quantify and adjust for residual cardiac-field leakage into the post-QRS EEG window, rather than relying on the QRS-exclusion window alone to guarantee a 'clean' interoceptive signal. Our CFA covariate (`cfa_r2_excl_qrs`, the fraction of post-QRS-window EEG variance explained by the concurrent ECG trace) is one defensible operationalisation of this contamination and is described fully in Methods; readers using a different CFA metric may find different residual associations.

Several simplifications were necessary to make this analysis tractable at this scale, and are worth stating plainly rather than glossing over. First, the mixed-effects model represents diagnosis as the same 3-level Neurological/Non-neurological/Unknown grouping used for the population-distribution analysis (Fig. 2), rather than as 15 non-exclusive binary diagnosis indicators; the latter would introduce substantial collinearity (most patients carry several comorbid categories) and would not fit legibly into a single coefficient plot. The grand-average waveform analysis (Fig. 4) instead examines individual diagnostic categories directly, providing a complementary, disease-specific view. Second, the 24-electrode montage was collapsed into 5 scalp regions for the mixed model to keep the fixed-effect count interpretable; full per-electrode descriptive statistics are retained in a supplementary table. Third, the diagnosis-vs-reference significance marks in Fig. 4 use the only comparison the existing cluster-permutation implementation supports -- an independent one-sample test (vs. zero) within each group -- rather than a genuine two-sample cluster statistic; a window flagged here means the diagnostic group shows a significant deflection from baseline that the reference cohort does not (or vice versa) at that stage, which is suggestive of a group difference but not a formal test of it. Future work extending the cluster-permutation framework to a true two-sample cluster test would sharpen this comparison.

Taken together, these results support the HEP as a physiologically meaningful, diagnosis-sensitive signal in a large clinical cohort, while highlighting cardiac field artifact as a confound that must be modelled explicitly, and motivate future work with two-sample cluster statistics, dense-electrode montages, and longitudinal or interventional designs to test whether HEP changes track disease progression or treatment response.

Methods

Cohort, EHR diagnosis categorisation, and cohort split

All data are from the Harvard Electroencephalography clinical polysomnography cohort processed by this repository's existing HEP pipeline (`6_hep_group_comparison.py`; `16_diagnosis_sleep_stage_comparison_dashboard.py`), reused here without modification via `mod16.load_patient_data` (`min_eeg_channels=None`, i.e. the full sparse+dense electrode pool) and the on-disk `individuals_cache*.pkl` produced by `get_group_individuals`. Diagnosis categories were derived from EHR diagnosis-name text (`bdsp_i0006_diagnosis` / `icd10_codes` parquet tables) via 15 non-exclusive, keyword-matched category rules (`_DIAG_CATEGORIES` in dashboard 16; identical rules were previously applied to build Paper CFA/demographics_combined.parquet, which we use directly as the primary

diagnosis-category and age/sex source, with `mod16.load_ehr_data / mod16._patient_demographics` as a fallback for the minority of patients not present in that cache). A patient was classed Neurological if they carried ≥ 1 of Cognitive Impairment/Dementia or Stroke/Cerebrovascular (regardless of comorbid non-neurological categories), Non-neurological if they carried ≥ 1 other category and zero neurological categories, and Unknown (reference) if they carried zero categories at all (`mod16.select_non_diagnosis_cohort` logic, applied here directly on the category list).

HEP epoching, preprocessing, and amplitude metric

HEP epochs were extracted by the repository's existing pipeline (`6_hep_group_comparison.process_file_data / get_group_individuals`), not reimplemented here: ECG is cleaned and R-peaks detected with a robust peak detector; EEG is epoched relative to each detected R-peak over a window of -0.3 to +0.4 s; epochs failing amplitude/flatline/roughness/spectral-power-ratio quality checks are excluded before averaging (`HEP_WINDOW_*` thresholds in `6_hep_group_comparison.py`); per-patient, per-electrode average HEP traces are the (`patient_id`, `hep_data`, `times`, `ch_names`, `rpeaks`, ...) tuples returned by `get_group_individuals` and cached to `individuals_cache*.pkl`. From each such trace we additionally required it pass `mod16._is_valid_hep_trace` (reject flatline, extreme-amplitude, or high-roughness traces) before inclusion in any analysis here. The scalar HEP amplitude metric used throughout (Figs. 2-3, and as the mixed-model outcome) is the mean of the (baseline-uncorrected) trace over 0.15-0.5 s post-R-peak (`AMP_WINDOW`, identical to `Paper1/build_diagnosis_alignment_analysis.py`), computed per electrode for the per-electrode long-format table and, separately, on the channel-averaged trace (`mod16._patient_hep_trace` over the montage-channel union) for the channel-averaged metric used in Figs. 2-3. Multiple recordings for the same canonical patient ID (`mod16._canonical_patient_id`, which collapses stage/session suffixes) within the same sleep stage were deduplicated to one recording per patient per stage.

Covariates: age, sex, heart rate, cardiac field artifact

Age and sex were taken from `Paper CFA/demographics_combined.parquet` where available, falling back to `mod16._patient_demographics` (COBRAD clinical sheet + EHR demographics cache) otherwise. Heart rate (mean ECG beats-per-minute, `qc_ecg_bpm`) and cardiac field artifact magnitude (`cfa_r2_excl_qrs`: the fraction of post-QRS-window EEG variance explained by the concurrent ECG trace, i.e. residual cardiac-field leakage into the analysis window after excluding the QRS complex itself) were taken from `Paper CFA/cfa_combined.parquet`, averaged per patient across all available recordings/channels/windows in that table (this table does not cover every HEP patient in the sparse-montage pool; the mixed-effects model is fit on the subset with complete age/sex/HR/CFA/region/stage/diagnosis data, N given in Results).

Cluster-based permutation testing

Group-level HEP significance was assessed with the repository's existing `permutation_cluster_jitter_test` (`6_hep_group_comparison.py`), used without modification: for a matrix of one channel-averaged trace per patient, the function computes a per-patient cluster permutation p-value (against pynapple-jittered circular-shift null traces), Fisher-combines these into a group-level p-value, and separately performs a group-level cluster-mass permutation test on the one-sample t-statistic (vs. zero) with the ± 50 ms window around the R-peak excluded from cluster detection (cardiac field artifact dominates this window). We ran this test independently, per sleep stage, on each diagnostic group's patient-average-HEP matrix and on the Unknown reference cohort's matrix (100 permutations, 0.1 s jitter). Because this function performs a one-sample (vs. zero) test rather than a two-sample group-difference test, we do not have a native two-sample cluster statistic available in this codebase; the diagnosis-associated windows shown in Fig. 4 are windows significant in the diagnostic group's one-sample test but not in the reference cohort's one-sample test at the same stage, which we report as a candidate/suggestive marker rather than a formal

group-difference test (see Discussion for this limitation).

Mixed-effects model

The mixed-effects model (Fig. 3) was fit with `statsmodels.formula.api.mixedlm`, REML, on the per-(patient, stage, electrode) long-format table restricted to rows with non-missing age, sex, heart rate, CFA, region, stage, and diagnosis group: `hep_amplitude_uv ~ C(neuro_group, Treatment('Unknown')) * C(stage, Treatment('light_sleep')) + C(region, Treatment('central')) + age_z + sex_bin + hr_z + cfa_z`, with a random intercept per patient (`groups=patient_id`). Age, heart rate, and CFA were z-scored; sex was coded Male=1, Female=0. Electrodes were mapped to 5 canonical scalp regions (frontal: Fp1/Fp2/F7/F3/Fz/F4/F8; central: C3/Cz/C4; parietal: P3/Pz/P4; temporal: T3/T7/T4/T8/T5/P7/T6/P8; occipital: O1/Oz/O2) to keep the electrode fixed effect legible in one coefficient plot, per the analysis plan; full per-electrode descriptive statistics are exported separately (Paper1/fig3_electrode_supplementary_table.csv). Diagnosis was represented as the same 3-level Neurological/Non-neurological/Unknown grouping used in Fig. 2, rather than 15 non-exclusive binary indicators, for the collinearity/interpretability reasons discussed above. We found that `statsmodels`' default `lbfgs` optimizer converged MixedLM to a degenerate boundary solution on this dataset (random-intercept variance forced to exactly zero, infinite log-likelihood, and grossly inflated covariate standard errors) -- confirmed on a 1,500-patient subsample where `lbfgs` alone diverged from `bfgs/cg/powell`, which agreed to 5 decimal places; we therefore fit the full model with the `bfgs` optimizer (`cg` as an automatic fallback if `bfgs` failed to converge). Approximate marginal/conditional R^2 were computed as $\text{variance-of-fitted-fixed-effects} / \text{total variance}$, and $(\text{variance-of-fitted-fixed-effects} + \text{random-intercept variance}) / \text{total variance}$ respectively, with $\text{total variance} = \text{fixed-effect variance} + \text{random-intercept variance} + \text{residual variance}$ -- an approximation of the Nakagawa & Schielzeth pseudo- R^2 rather than an exact computation.

Statistics summary

Group comparisons of the population HEP-amplitude distribution (Fig. 2) used the Kruskal-Wallis H test across the three diagnosis groups, per sleep stage, with Benjamini-Hochberg FDR correction across stages (`mod16.benjamini_hochberg`). All p-values reported are two-sided. Analyses were run with the project's existing `venv` (Python 3.11; `statsmodels`, `scipy`, `matplotlib`, `reportlab`); no raw EDF/EEG data were reprocessed for this manuscript beyond what the existing pipeline scripts already cache on disk.

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